



PROJECT REPORT

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Introduction & Problem

Motivation:-

Verifying Character Backstory in Long Novels

Imagine reading a **100,000+ word novel** and being asked: “Does this character’s backstory actually fit the story?”

It’s not simple. Critical clues can be **hundreds of pages apart**. Maybe the character does the impossible, or past events make actions irreversible. To detect inconsistencies, you need **temporal reasoning, causal logic, and evidence aggregation**—plausibility isn’t enough.

LLMs are impressive, but they struggle here:

- **Limited context windows** → key facts can be lost
- **Probabilistic outputs** → inconsistent, hard-to-explain decisions
- **Blended reasoning and generation** → convincing but potentially ungrounded explanations

In fields like law, compliance, and enterprise knowledge, conclusions must be traceable, provable, and reproducible. Contradictions must be demonstrable, not inferred.

Our solution: a deterministic reasoning system that:

- Preserves temporal structure
- Checks symbolic consistency
- Uses LLMs only as auxiliary tools (event detection, claim normalization)

By **separating perception from reasoning** and enforcing **explicit rules**, the system reliably evaluates long narratives, catching subtle inconsistencies **without relying on LLMs to decide truth**.

Task Definition & Evaluation Setting

Objective

Determine whether a character's hypothetical backstory is logically and causally consistent with the events of a full-length novel.

Input

1. Full Novel Text

- Complete long-form narrative (**100,000+ words**) as plain text
- Must be ingested entirely, without truncation or summary
- Relevant evidence may appear in distant, non-contiguous parts, requiring temporal order preservation and global context

2. Hypothetical Character Backstory

- Separately written description of one central character
- Includes early-life events, formative experiences, beliefs, motivations, fears, assumptions
- Crafted to appear plausible but may contain subtle or deep inconsistencies
- Must be evaluated against the entire novel, not just a local context

Output

1. Binary Consistency Label

- **1** = Backstory is **consistent**
- **0** = Backstory is **contradict**

2. Optional Evidence-Based Rationale

- Concise explanation for consistency or contradiction
- If contradictory, rationale should reference:
 - Which backstory claim was evaluated
 - Which narrative constraint was violated
 - Evidence from the novel supporting the decision

Rationale

The rationale provides transparency and debugging, but the primary evaluation criterion is the correctness of the consistency label.

Evaluation Setting

- This is not a traditional machine learning problem.
- No training, fine-tuning, or parameter optimization is performed.
- No learned weights or statistical fitting—the system relies on deterministic logic and symbolic constraints.
- The provided training set is used only for sanity-checking reasoning logic, ensuring:
 - Constraints are correctly enforced
 - Temporal order is respected
 - Obvious contradictions are detected
- No information from training data influences inference on the test set.

Role of Test Data

- The test set is used only for final inference.
- No inspection, tuning, or rule changes are made after observing test results.
- Each instance is processed independently using the same fixed reasoning pipeline.

Problem Framing Summary

- This is framed as a logical consistency checking problem over long contexts, not as:
 - **Text generation**
 - **Summarization**
 - **Question answering**
 - **Entailment classification**
- **Key difficulty:** aggregating evidence across long narratives, respecting temporal order, and enforcing causal and narrative constraints.
- **The system decides whether a backstory is logically possible given the novel, not just whether it sounds plausible.**

Design Principles & System Constraints

System Design for Long-Context Consistency Checking

The system ensures reliable narrative consistency under strict reasoning constraints. It follows a unified philosophy and supports **two approaches**, sharing core logic and constraints, differing only in how facts and claims are extracted.

3.1 Shared Design Principles

- **Reasoning Over Generation:** Decisions based on symbolic rules, not LLM outputs
- **Deterministic Final Decision:** Same input → same output; no randomness
- **Time-Aware Evidence Aggregation:** Events temporally ordered, irreversible constraints enforced
- **No Truncation/Summarization:** Entire novels ingested; chunking only for efficiency
- **Explainability Over Hacks:** Every decision traceable to evidence and rules

Ensures correctness, reproducibility, and alignment with the problem.

3.2 Approach 1: Fully Rule-Based Deterministic Reasoning

- **No LLMs used**
- Character states extracted via rule patterns
- Backstory claims identified using lexical patterns
- Consistency checks done **with symbolic constraints**

Guarantees:

- Deterministic, fully explainable reasoning
- Strict temporal and causal adherence
- No probabilistic outputs

3.3 Approach 2: LLM-Assisted Perception with Deterministic

Reasoning

- **Augments reasoning with LLMs for perception-level tasks only:**
 - Event classification
 - Claim normalization
 - Semantic disambiguation
- **LLMs are never allowed to:**
 - Decide contradictions
 - Compare evidence
 - Assign final labels
- **All final decisions use the same symbolic constraint engine as Approach 1**

Benefits:

- Improves recall for implicit or ambiguous events
- Maintains deterministic outputs and explainable reasoning
- Enforces logical correctness even if LLM outputs are noisy

Demonstrates a reasoning-first architecture with optional semantic assistance—not an LLM-based solution.

System Architecture Overview

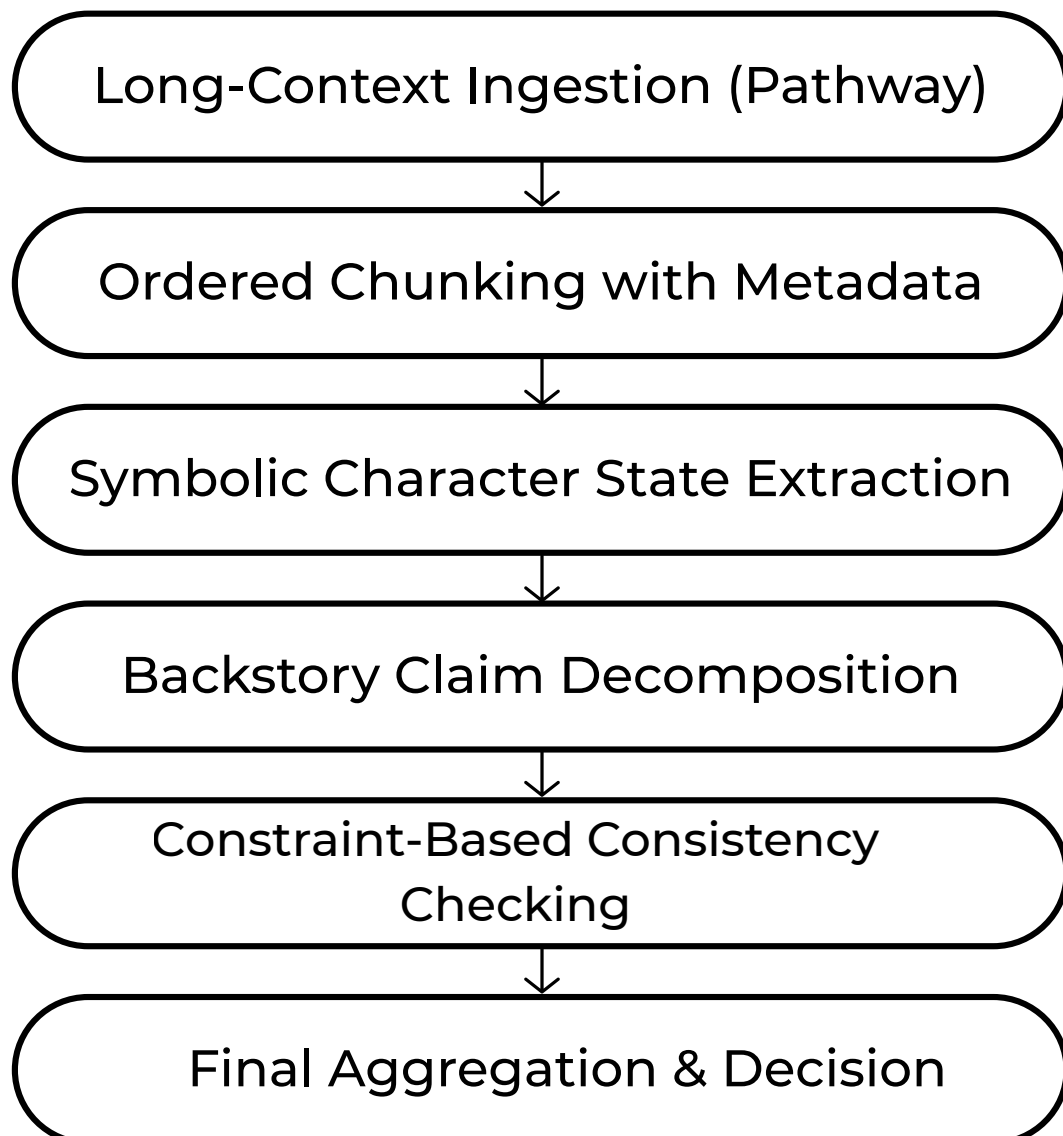
System Architecture: Single Deterministic Reasoning Pipeline

The system is a single deterministic architecture for long-context narrative consistency checking.

- Core reasoning pipeline is identical across experiments.
- Supports two perception configurations (Approach 1 & 2), differing only in how raw text is converted into symbolic inputs.
- All reasoning, constraint enforcement, and final decisions are deterministic and shared.

4.1 High-Level Pipeline

The system follows a linear, staged pipeline, where each stage has a clear responsibility:



Each stage operates independently and passes structured outputs to the next.

Ensures modularity, traceability, and reproducibility.

4.2 Long-Context Ingestion (Pathway)

- Used exclusively for infrastructure-level ingestion and orchestration of long documents
- Entire novels ingested without truncation or summarization
- Handles texts **100,000+ words** in full
- Manages scalable loading, indexing, and ordered storage
- No semantic reasoning occurs at this stage
- Preserves global narrative context, enabling reasoning across arbitrarily long timelines.

4.3 Ordered Chunking with Metadata

- Novels are divided into fixed-size, overlapping chunks for efficient processing.
- Each chunk has a positional index reflecting its order in the story.
- Metadata includes book identity and narrative position.
- Chunking is for memory and processing efficiency only, not semantic reduction.
- Temporal order is preserved throughout the pipeline.

4.4 Symbolic Character State Extraction (Perception Layer)

- Converts narrative text into symbolic character facts, including:
 - Death events
 - Imprisonment (start–end ranges)
 - Escape events ending imprisonment
 - Other irreversible narrative states

- **Facts are stored as symbolic constraints, indexed by:**
 - **Character identity**
 - **Temporal position**
- Once stored, these constraints are treated as irreversible system truths

Perception Configuration:

- **Approach 1 (Rule-Only):** Character states extracted via deterministic pattern rules, no LLMs.
- **Approach 2 (LLM-Assisted):** LLMs assist only in detecting implicit or ambiguous events (e.g., indirect death references); all facts pass through symbolic constraint memory.

In both approaches, no LLM performs reasoning or validation.

Long-Context Handling with Pathway

Pathway is used as an infrastructure layer, not a reasoning engine. It enables:

- Full document ingestion without truncation
- Ordered, metadata-aware storage
- Scalable, deterministic orchestration

This avoids the limitations of naïve RAG systems that rely on top-k retrieval and implicit LLM judgment.

Key Guarantees:

- Entire novels ingested end-to-end
- Chunking preserves order and metadata
- No summarisation
- No reasoning inside Pathway

Pathway ensures global narrative access while keeping reasoning explicit and symbolic.

Symbolic Character States & Constraints

7.1 Symbolic State Representation

Narrative events are converted into symbolic character states, such as:

- **Death**
- **Imprisonment (start–end interval)**
- **Restricted movement**

Each state is indexed by:

- Character identity
- Temporal position

7.2 Irreversible Constraints

Once established, constraints cannot be overridden.

Example

If **death_time(character) = T**

Then any action at time $> T \rightarrow$ CONTRADICTION

This enforces causal logic mechanically, not probabilistically.

7.3 Fact vs Interpretation

- Facts $\rightarrow$ extracted from text, stored permanently
- Interpretations $\rightarrow$ handled only by deterministic logic

LLMs may assist in detecting facts, but never interpret them.

Backstory Claim Decomposition & Temporal Modelling

Backstories are decomposed into atomic claims, each representing:

- **One character**
- **One action or state**
- **One temporal position**

Example:

“After escaping prison, John later fights in the war.”

→

- John escapes imprisonment
- John participates in combat

Temporal cues (“after”, “during”, “later”) determine ordering, not validity.

Backstories are treated as hypothetical futures that must conform to fixed narrative constraints.

Deterministic Consistency Checking Logic

This is the system's authoritative reasoning layer.

Core Rules:

- Death Rule: No actions after death
- Imprisonment Rule: No free actions during confinement
- Escape Rule: Terminates imprisonment interval

Evaluation Process

1. Iterate through atomic claims
2. Retrieve character constraints
3. Compare temporal positions
4. Apply explicit rules
5. Aggregate violations

Decision Rule

- Any violation → **0 (Contradiction)**
- No violations → **1 (Consistent)**

No scoring, voting, thresholds, or LLM judgment.

Limitations & Future Extensions

Limitations:

- **Implicit Narrative Events:** Indirect events (e.g., hints of death/disappearance) may evade rule-based detection.
- **Figurative Language:** Metaphors or sarcasm (e.g., “dead to the world”) can be mistaken for factual events.
- **Alias & Identity Ambiguity:** Characters under multiple names can fragment state tracking; system assumes consistent naming.

Future Extensions:

- **Negative & Soft Constraints:** Flag highly implausible claims while keeping deterministic decisions.
- **Counterfactual Consistency Testing:** Explore hypothetical alternative timelines for backstory evaluation.
- **Character Evolution Modeling:** Track long-term roles, affiliations, moral shifts to detect higher-order inconsistencies.

Most failures come from linguistic ambiguity, not missing data. By prioritizing explainable, irreversible logic, the system ensures defensible and traceable decisions over fragile generative plausibility.

Real-World & Business Applications

Although tested on novels, the system applies to any long, evolving documents where consistency over time is critical. In enterprise and institutional contexts, plausibility or local context is insufficient; deterministic, time-aware reasoning over full histories is required.

Applicable Domains:

- **Legal & Contract Auditing:** Detects contradictions between amendments and earlier obligations using irreversible constraints.
- **Compliance & Policy Drift:** Flags actions invalidated by policy updates, enforcing temporal constraints.
- **Enterprise Knowledge Bases:** Ensures explanations and documentation remain consistent with authoritative rules.
- **Financial Due Diligence:** Identifies inconsistencies in investment theses caused by temporal misalignment in filings.

Role of Pathway:

- Enables full ingestion of large document collections without truncation
- Preserves temporal order and metadata
- Orchestrates deterministic, auditable reasoning pipelines

Summary:

By treating novels as stand-ins for complex documents, the system demonstrates that deterministic, constraint-based long-context reasoning—powered by Pathway—is a general-purpose capability for legal, compliance, enterprise, and financial domains.

Conclusion

This work presents a deterministic system for verifying narrative consistency over long-form texts, using full novels as a proxy for real-world long-context reasoning challenges. The task is reframed from language generation or entailment to logical consistency checking over time-indexed symbolic facts.

The key contribution is a strict separation of perception and reasoning: LLMs are used only to extract structured signals from text, while all final decisions are made through explicit, rule-based constraints. Narrative events are converted into irreversible symbolic states, enabling deterministic and explainable evaluation of hypothetical backstories.

Long-context handling is treated as a first-class requirement. Using Pathway, the system ingests entire documents without truncation, preserves temporal order, and supports reasoning over evidence distributed across long timelines—addressing core limitations of end-to-end LLM approaches.

The result is a system that is auditable, reproducible, and logically grounded, making it applicable beyond literature to high-stakes domains such as legal analysis, compliance, and enterprise knowledge validation. More broadly, this work demonstrates that reliable long-context reasoning requires symbolic memory, temporal structure, and deterministic constraint enforcement—not larger language models alone.